

# Power Query Transformations Guide

## Coding-Forge | Cleaning Financial Data in Power BI

### Introduction

Real-world data from Dynamics 365 and Dataverse exports is never clean. This guide walks you through fixing common data quality issues using only the Power Query Editor ribbon buttons — no code required.

### How to Open Power Query Editor

1. In Power BI Desktop, go to Home > Transform Data
2. The Power Query Editor window opens
3. Each table appears on the left under "Queries"
4. Your cleanup steps are recorded on the right panel under Applied Steps

Tip: Turn on Column Quality, Column Distribution, and Column Profile from the View tab. These show data issues at a glance.

### File 1: Budget vs Actuals

#### What's Wrong With This Data?

| Problem                          | What It Looks Like                                         |
|----------------------------------|------------------------------------------------------------|
| Dollar signs in numbers          | \$150000, -\$120000                                        |
| Percentage signs                 | 6.0%                                                       |
| Inconsistent capitalization      | materials, travel, OVER BUDGET, at risk                    |
| Extra spaces                     | " Cyber Defense Initiative"                                |
| N/A text in number columns       | N/A                                                        |
| Footer/notes rows at the bottom  | "Notes: Data exported from..." + blank row + duplicate row |
| Shortened names                  | "Satellite Comm Upgrade" instead of full name              |
| Multiple fiscal years & quarters | FY2025 (Q1–Q4) and FY2026 (Q1–Q2)                          |

#### How to Fix It

##### 1. Remove the footer/notes rows at the bottom

- Home > Remove Rows > Remove Bottom Rows > Type 3
- (This removes the blank row, notes row, and duplicate row at the bottom in one step)

##### 2. Trim extra spaces from text

- Select all text columns (hold Ctrl and click: Program Name, Program ID, Cost Center, Budget Category, Funding Source, Status)
- Transform > Format > Trim

### 3. Fix dollar signs and commas in amount columns

- Select **Budgeted Amount**, **Actual Amount**, and **Variance** columns (hold Ctrl to multi-select)
- Transform > Replace Values > Find: \$ Replace with: (leave blank) > OK
- Transform > Replace Values > Find: , Replace with: (leave blank) > OK

### 4. Fix percentage signs

- Select **Variance %** column
- Transform > Replace Values > Find: % Replace with: (leave blank) > OK

### 5. Fix N/A values

- Select **Actual Amount**, **Variance**, and **Variance %** columns
- Transform > Replace Values > Find: N/A Replace with: (leave blank) > OK

### 6. Change columns to the right data type

- Click the "ABC" icon on **Budgeted Amount** column header > Select Decimal Number
- Click the "ABC" icon on **Actual Amount** column header > Select Decimal Number
- Click the "ABC" icon on **Variance** column header > Select Decimal Number
- Click the "ABC" icon on **Variance %** column header > Select Decimal Number

### 7. Fix inconsistent capitalization in Budget Category

- Select **Budget Category** column
- Transform > Format > Capitalize Each Word

### 8. Fix "Sub-Contracts" → "Subcontracts"

- Select **Budget Category** column
- Transform > Replace Values > Find: Sub-Contracts Replace with: Subcontracts

### 9. Fix Status column consistency

- Select **Status** column
- Transform > Format > Capitalize Each Word

### 10. Fix shortened program names

- Select **Program Name** column
- Transform > Replace Values:
- Find: **Satellite Comm Upgrade** Replace with: **Satellite Communications Upgrade**
- Find: **Next Gen Radar Systems** Replace with: **Next-Gen Radar Systems**

## File 2: Purchase Orders

### What's Wrong With This Data?

| Problem                   | What It Looks Like                              |
|---------------------------|-------------------------------------------------|
| Different date formats    | 10/15/2024, 11-20-2024, 2025-02-01, Feb 10 2025 |
| Dollar signs and commas   | "\$1,250,000.00"                                |
| NULL and dashes as values | NULL, -                                         |

| Problem                              | What It Looks Like                            |
|--------------------------------------|-----------------------------------------------|
| Inconsistent status capitalization   | CLOSED, closed, in progress                   |
| Inconsistent priority capitalization | CRITICAL, medium, high                        |
| Extra spaces in names                | "Triton Systems ", " James Mitchell"          |
| Vendor name variations               | "Apex Dynamics Corp" vs "Apex Dynamics Corp." |
| Duplicate row                        | PO-1006 appears twice                         |
| Shortened program name               | "Satellite Comm Upgrade"                      |

## How to Fix It

### 1. Remove duplicate rows

- Select the **PO ID** column > Home > Remove Rows > Remove Duplicates

### 2. Trim spaces from all text columns

- Select all text columns > Transform > Format > Trim

### 3. Clean the dollar amounts

- Select **PO Amount** column
- Transform > Replace Values: Find \$ Replace with blank
- Transform > Replace Values: Find , Replace with blank
- Repeat for **Amount Paid** and **Amount Remaining**

### 4. Fix NULL and dash values

- Select **Amount Paid** column
- Transform > Replace Values: Find NULL Replace with 0
- Transform > Replace Values: Find - Replace with 0

### 5. Change amount columns to Decimal Number type

- Click column type icon on **PO Amount** > Decimal Number
- Repeat for **Amount Paid** and **Amount Remaining**

### 6. Fix date columns

- Select **Order Date** column > Click the type icon > Choose Date
- Power BI will auto-detect most formats
- If any rows show "Error", right-click the column > Replace Errors > enter null
- Repeat for **Delivery Date**

### 7. Fix PO Status capitalization

- Select **PO Status** column
- Transform > Format > Capitalize Each Word

### 8. Fix Priority capitalization

- Select **Priority** column
- Transform > Format > Capitalize Each Word

### 9. Fix vendor name inconsistencies

- Select **Vendor Name** column
- Transform > Replace Values: Find **Apex Dynamics Corp.** Replace with **Apex Dynamics Corp**
- Transform > Replace Values: Find **Centurion Inc.** Replace with **Centurion Inc**

## 10. Fix shortened program names

- Select **Program Name** column
  - Transform > Replace Values: Find **Satellite Comm Upgrade** Replace with **Satellite Communications Upgrade**
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# File 3: Programs (Dimension Table)

## What's Wrong With This Data?

Nothing! This is a clean lookup table with one row per program. It serves as the central dimension for the star schema.

## Steps

### 1. Verify data types

- All columns should import as Text — no changes needed

### 2. Rename the query

- Right-click the query name on the left > Rename > **Programs**
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## Tips for the Exercises

## How to Know You're Done Cleaning

Use the View tab and enable:

- Column Quality — shows % valid, error, and empty per column
- Column Distribution — shows distinct/unique value counts
- Column Profile — shows statistics for the selected column

You want:

- 0% errors in all columns
  - Number columns showing as numbers (not text)
  - Consistent values (check Column Distribution — you shouldn't see **Closed** AND **closed**)
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## Common Mistakes to Avoid

1. Don't forget to click "Close & Apply" when done — otherwise nothing saves
  2. Order matters — Remove rows FIRST, then fix values (don't waste time cleaning rows you'll delete)
  3. Watch for partial matches — When replacing text, make sure you don't accidentally create doubled words
  4. Check date conversions — After changing to Date type, look for "Error" values that didn't convert
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## Undo a Step

If you make a mistake:

- Look at Applied Steps on the right panel

- Click the X next to any step to delete it
  - Steps below it will also be removed
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## After Cleanup: Setting Up Relationships

Once all 3 tables are clean and loaded:

1. Go to Model View (left sidebar)
2. Drag **Program ID** from Programs to **Program ID** in Budget vs Actuals
3. Double-click the relationship line to open the dialog:
  - Cardinality: One-to-Many
  - Cross filter direction: Single
4. Drag **Program ID** from Programs to **Program ID** in Purchase Orders
5. Double-click the relationship line:
  - Cardinality: One-to-Many
  - Cross filter direction: Single

Why Single? In a star schema, filters flow from the dimension table (Programs) down to the fact tables. A slicer on Program Name will filter both Budget vs Actuals and Purchase Orders automatically. "Both" is generally not recommended — it can cause ambiguity and performance issues.

The Programs table is your dimension table (star schema center). Both Budget vs Actuals and Purchase Orders are fact tables that connect to it. This enables cross-filtering between all tables through the shared Program dimension.

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Next: See `REPORT_BUILDING_GUIDE.md` for how to build each report.